

Deepfake Detection Challenge Student Handout

TOK Cognitive Science Activity Bank • Knowledge and Technology • 25-35 min

Category	Knowledge and Technology	Time	25-35 min
Difficulty	Medium	ID	deepfake-detection-challenge

Big idea: Evidence standards change when media is synthetic.

Student-Facing Prompt

Inspect Deepfake Detection Challenge Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Deepfake Detection Challenge Stimulus A	Student-facing stimulus 1: Three image descriptions: authentic-looking, edited-looking, and ambiguous. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Deepfake Detection Challenge Stimulus B	Student-facing stimulus 2: A checklist for source, context, lighting, body geometry, and independent corroboration. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Use this for comparison, a second round, or a partner challenge.
Deepfake Detection Challenge Reveal / Twist	Student-facing stimulus 3: A debrief comparing technical clues with social trust cues. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Teacher reveal: connect the changed response to the big idea — Evidence standards change when media is synthetic.

Actual Lesson Questions

#	Question for students
1	After inspecting Deepfake Detection Challenge Stimulus A, what is your first knowledge claim?
2	Which exact detail from Deepfake Detection Challenge Stimulus A did you use as evidence?
3	What did you infer beyond the evidence?
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?
5	What missing information would most improve the knowledge claim?
6	What happens to knowledge when evidence can be easily fabricated?

Ready-to-Use Examples

- Three image descriptions: authentic-looking, edited-looking, and ambiguous.
- A checklist for source, context, lighting, body geometry, and independent corroboration.
- A debrief comparing technical clues with social trust cues.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Knowledge and Technology.

Activity Steps

1. Show several media examples without revealing which are manipulated.
2. Students create a checklist: source, metadata, shadows, hands, audio mismatch, reverse image search, corroboration.
3. Reveal answers or uncertainty.
4. Students discuss how much trust should change in a synthetic media environment.

Observation and Evidence Record

Use this grid to keep observation, inference, evidence, and confidence separate.

What I noticed	What I inferred	Evidence	Confidence

TOK Debrief

- Knowledge question: What happens to knowledge when evidence can be easily fabricated?
- Do new technologies require new standards of proof?
- Is "seeing is believing" still defensible?

Knowledge Questions

- What happens to knowledge when evidence can be easily fabricated?
- Do new technologies require new standards of proof?
- Is "seeing is believing" still defensible?

Transfer Example

For example, run this activity with three image descriptions: authentic-looking, edited-looking, and ambiguous. Have students commit to an initial claim, then reveal the change, hidden rule, alternative context, or comparison that makes the knowledge problem visible.

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...